

Correct review standard for this proof class

This dossier accompanies a constraint-formalism impossibility proof. It should not be reviewed as a claimed PDE-native continuation-estimate proof. Clay's rules require a complete mathematical solution of the official problem description after publication and public examination; they do not require a preferred analytic proof template. The decisive review questions are: ambient proof-system admissibility as kernel-role instantiation, counterexamplehood as theorem-bearing construction, kernel necessity for non-degenerate regimes, official periodic Navier–Stokes proof/refutation-regime non-degeneracy, finite-horizon endpoint-condition induction, endpoint-status exhaustion, and counterexample soundness. Method-neutrality is not used to reduce the mathematical burden: the submission still claims to prove Fefferman's periodic statement (B), but by impossibility of same-scope counterexample completion rather than by an analytic continuation estimate.

Supplementary Referee Dossier

Proof-Class and Source-Control Audit

Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations

Refined manuscript + AASC corpus-matrix controlled refinement pass

Audited entry files.

Main manuscript: `main.tex`

Readiness companion: `phase7_readiness_pack.tex`

Supplementary dossier: `supplementary_referee_dossier.tex`

Manuscript state audited here: **closure audit refined packet + controlled AASC corpus-matrix refinement pass.**

Purpose. This dossier is a referee-facing audit aid. It does not add proof burden, new theorems, or new primitives. Its job is to state the current live proof spine, identify the exact role of Lemma 12F in the refined submission, and record the additional fixed-domain, continuation-locus, metric non-authority, claim-status, and role-cardinality control layers added by the corpus-matrix pass.

Executive verdict. In the current packet, 12F is still *live*, but it is no longer presented as the whole exclusion engine. The manuscript states on the page that 12F is a fixed-periodic scope-fixity / typing premise only, that 12G is the first node where P4 yields the no-second-gate consequence, and that later nodes carry this packet forward in one direction only. The honest defense posture is therefore: the 12F seam is **NARROW**, not magically gone. Keeping the objection alive requires denying either 12F as a legitimate bridge premise, 12G/P4 as the first exclusion node, or the downstream transport chain through 13K / 20.5B / 20B and ST2–ST3 / 20C.

Scope of this dossier.

- It audits the manuscript as currently written.
- It does *not* argue Clay eligibility, publication acceptance, or community acceptance.
- It does *not* replace the manuscript with philosophy or side papers.
- It records where the bridge has been clarified and where a skeptical reader may still press.

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1. Reader orientation spine

The live proof route is: (1) kernel necessity for non-degenerate theorem-bearing proof/refutation regimes; (2) ambient counterexamplehood as theorem-bearing construction; (3) Fefferman proof/refutation-regime non-degeneracy; (4) finite-horizon endpoint-condition induction; (5) Counterexample Soundness; (6) endpoint-status exhaustion; and (7) branch elimination and official endpoint compression. The full dossier below is an audit and denial-target map for that spine; it does not add proof burden.

2. Controlled corpus-matrix support additions

Matrix status. The uploaded AASC corpus matrix is used as a governance and refinement-control ledger only. It is not a live proof source. The controlled pass internalizes only local material that is restated in the manuscript, and leaves broader corpus titles as provenance, audit-only, or quarantined support.

Refinement layer	Manuscript insertion	Referee-facing gain
Fixed-domain comparison class	Definitions 3.4A–3.4B and the fixed-domain criterion audit around 13K.1	A counterexample must preserve the official periodic comparison class before it can be a same-scope objection.
Continuation-locus / persistence regime	Definitions 18.0A–18.0C and lemmas 19E.1–19E.3	The realized maximal classical lineage is treated as one persistence regime rather than time-indexed admissibility slices.
Metric non-authority	Theorem 13.23A.6 and the BKM appendix refinement	BKM, critical norms, profile growth, and Θ_x -style data are bookkeeping or fixed-witness packaging unless promoted illicitly into a second gate.
Role-cardinality endpoint audit	Appendix F role-cardinality audit	Endpoint closure is framed as failure of a compatible finite-horizon singular endpoint role, not as an unconstrained singleton assertion.
Claim-status warrant	Appendix F claim-status table	The proof class, target carrier, non-claims, and forbidden laundering channels are visible in one audit object.

3. Current live proof spine

Live route only.

- (S1) **Fixed-scope bridge.** The Section 12 packet culminates in 12G, where P4 is applied at the fixed periodic scope after 12C–12F have supplied the bridge hypotheses.
- (S2) **Carrier packet.** 13K carries the no-new-criterion consequence into the periodic carrier packet, while 20.5B and 20B transport that same consequence into the realized classical layer.
- (S3) **Standing packet.** ST2 and ST3 carry the same fixed-scope packet into standing / admissibility language; 20C.1 carries it into endpoint-status language.
- (S4) **Realized-lineage packet.** 19C–19F derive PDE-native admissibility on the maximal classical interval; 20B.1–20B.3 and ST4–ST5 bind the admitted carrier seed to one realized lineage and block time-indexed admissibility dynamics.
- (S5) **Endpoint collapse.** 20C.1–20C.3 state and collapse the realized-lineage status trichotomy; 21.10 / 21.15 / 21A / 21.17 compress the final endpoint.

4. Bridge-dependency refinements

Phase	Patch target	Referee-facing effect
0	Front-matter governance	Added method-neutral Clay-rules paragraph, no-hidden-premise sentence, and one live endgame spine.
1	12F role clarification	12F now says it is same-scope typing / scope-fixity only; added the two-case split and the “What 12F is not” box.
2	12G / P4 bridge tightening	12G states explicitly that 12F contributes boundary-signature closure / scope-fixity and that P4 is the first place the no-second-gate consequence enters.
3	Downstream dependency transparency	13K, 20.5B, and 20B now say they inherit the exclusion consequence from 12G / P4 rather than from 12F in isolation.
4	Standing-chain clarification	ST2, ST3, and 20C.1 now state that the fixed-scope standing packet is transported from 12G / P4, not restarted from 12F.
5	Endgame visibility + audit	Page 1 and Appendix F now expose the unique bridge-to-endgame chain in one place.
6	Lexical / referee pass	Compressed summaries now all say the same thing: 12F is a live premise only as scope-fixity, 12G / P4 is the first exclusion node, and the external referee package uses that same narrow honest posture.

5. The 12F seam: exact current position

5.1 What the manuscript now claims

- 12F is **not** a PDE estimate.
- 12F is **not** a continuation criterion.
- 12F is **not** a standalone impossibility theorem.
- 12F **is** the fixed-periodic typing / scope-fixity form of boundary-signature closure.
- 12G is the first node where P4 is applied to that completed packet and where the no-second-gate consequence is locally obtained.

5.2 What remains live

12F is still on the live route. The manuscript does *not* now claim that 12F is dispensable. Instead it claims a narrower and more honest ordering:

12F (scope-fixity premise) $\rightarrow$ 12G/P4 (first exclusion node) $\rightarrow$ downstream carry-through

5.3 What a skeptical referee must now deny

To keep the current 12F objection alive, a referee must deny at least one of the following:

- that 12F is a legitimate fixed-periodic scope-fixity / typing premise;
- that 12G validly applies P4 to the completed periodic bridge packet;
- that 13K / 20.5B / 20B correctly carry the no-new-criterion consequence forward;
- that ST2–ST3 / 20C.1 / 20C correctly carry the same fixed-scope packet into standing / endpoint-status language.

That is materially narrower than the earlier blanket complaint that “12F is circular, therefore everything downstream fails.”

6. Current denial-by-theorem map

Objection	To keep it alive, deny at least one of	Status
12F is still the whole no-second-gate engine.	12F as a legitimate fixed-scope typing premise; 12G/P4 as first exclusion node; downstream carry-through via 13K / 20.5B / 20B and ST2–ST3 / 20C.1 / 20C.	NARROW
The realized lineage can be re-classified slice-by-slice.	20B.1, 20B.2, 20B.3, ST4, ST5.	CLOSED

Objection	To keep it alive, deny at least one of	Status
There is still a neutral third same-scope endpoint status.	20C.1, 20C.2, 20C, 20C.3.	CLOSED
A hidden same-scope discriminator can enter through history, order, or representative-only diagnostics.	Standing quotient theorem, factorization discipline, history-factorization theorem, ATS role packet, skin non-authority corollaries.	CLOSED
Metric divergence itself is the singularity gate.	Metric / tensor / witness localization packet plus 20B / 20C.	CLOSED
The paper claims admissibility supplies the missing analytic bridge inequality.	Front-matter claim-type box, non-claim list, no-new-criterion theorems.	CLOSED

7. Referee guidance

How to use this dossier.

- (1) Read page 1 of the manuscript first. The bridge-seam live dependency chain is now there explicitly.
- (2) If you think 12F is still circular, identify whether your complaint attacks: 12F as premise, 12G/P4 as bridge application, or one of the downstream transport nodes.
- (3) Do not report the objection in the vague form “this feels circular.” The stress-test protocol requires a theorem-local target.
- (4) Use Appendix F for the node-by-node bridge audit, not as proof support but as a navigation aid.

Bottom line. The bridge-seam refinement does not change the proof class. It changes the visibility and asymmetry of the bridge. The seam is tighter, more honest, and more referee-readable; it is not claimed to be metaphysically invisible.

Correct review standard after Lean / kernel clarification

The submission should not be rejected merely for lacking a PDE-native continuation estimate. Its claim class is a constraint-formalism impossibility proof. The public Lean repository certifies the manuscript-facing structural surface and axiom-audit status; the manuscript then proves that the official periodic Navier–Stokes regime is non-degenerate, so the fixed-scope kernel applies by necessity. A substantive objection must deny kernel necessity, official-regime non-degeneracy, finite-horizon endpoint-condition induction, counterexample soundness, endpoint-status exhaustion, or branch elimination.